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1. Add two numbers

Welcome | add.c | oddeven.c

add.c > main()

```c
#include <stdio.h>
int main(){
    int num1, num2, sum;
    printf("Enter the first number: ");
    scanf("%d", &num1);
    printf("Enter the second number: ");
    scanf("%d", &num2);
    sum = num1 + num2;
    printf("The sum of the given number is: %d", sum);

}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZUR

```
cd "/Users/adarshaprasai/Documents/Herald/Final Year/Sem1/HPC/B
./"add"
adarshaprasai@Adarshas-MacBook-Air Bonusclass_cw % cd "/Users/a
adarshaprasai@Adarshas-MacBook-Air output % ./"add"
Enter the first number: 101
Enter the second number: 102
The sum of the given number is: 203%
adarshaprasai@Adarshas-MacBook-Air output %
```

2. Check if a number is odd or even

Welcome | add.c | oddeven.c

oddeven.c > main()

```c
#include <stdio.h>
int main (){
    int num1;
    printf("Enter a number: ");
    scanf("%d", &num1);
    if (num1%2==0)
    {
        printf("%d is even.", num1);
    }
    else{
        printf("%d is odd.", num1);
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
cd "/Users/adarshaprasai/Documents/Herald/Final Year/Sem1/H
./"oddeven"
adarshaprasai@Adarshas-MacBook-Air Bonusclass_cw % cd "/Use
adarshaprasai@Adarshas-MacBook-Air output % ./"oddeven"
Enter a number: 10311
10311 is odd.%
adarshaprasai@Adarshas-MacBook-Air output %
```

3. Find the largest of three numbers

largestofthree.c > main()

```c
#include <stdio.h>
int main (){
    int num1, num2, num3, largest;
    printf("Enter number1: \n");
    scanf("%d", &num1);
    printf("Enter number2: \n");
    scanf("%d", &num2);
    printf("ENter number3: \n");
    scanf("%d", &num3);
    largest = num1;
    if(num2 > largest){
        largest = num2;
    }
    if(num3 > largest){
        largest = num3;
    }
    printf("The largest number is: %d", largest);
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE

```
cd "/Users/adarshaprasai/Documents/Herald/Final Year/Sem1/HPC/Bonuscla
./"largestofthree"
adarshaprasai@Adarshas-MacBook-Air Bonusclass_cw % cd "/Users/adarshap
adarshaprasai@Adarshas-MacBook-Air output % ./"largestofthree"
Enter number1:
10
Enter number2:
11
ENter number3:
3
The largest number is: 11%
```

4. Check if a year is a leap year

leapyear.c > main()

```c
#include <stdio.h>
int main (){
    int year;
    printf("Enter a year: \n");
    scanf("%d", &year);
    if(year%4==0){
        printf(" Year %d is a leap year", year);
    }
    else{
        printf("Year %d is not a leap year", year);
    }

}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE

```
cd "/Users/adarshaprasai/Documents/Herald/Final Year/Sem1/HPC/Bonusclass_cw/output"
./"largestofthree"
adarshaprasai@Adarshas-MacBook-Air Bonusclass_cw % cd "/Users/adarshaprasai/Documer
adarshaprasai@Adarshas-MacBook-Air output % ./"largestofthree"
Enter number1:
10
Enter number2:
11
ENter number3:
3
The largest number is: 11%
adarshaprasai@Adarshas-MacBook-Air output % cd "/Users/adarshaprasai/Documents/Hera
./"leapyear"
adarshaprasai@Adarshas-MacBook-Air output % ./"leapyear"
Enter a year:
2020
 Year 2020 is a leap year%
adarshaprasai@Adarshas-MacBook-Air output %
```

5. Check if a character is a vowel or consonant

leapyear.c | largestofthree.c | vowelornot.c | sum_array.c | string_len.c

vowelornot.c > main()

```c
#include <stdio.h>
int main (){
    char ch;
    printf("Enter a character: \n");
    scanf("%c", &ch);
switch (ch)
{
case 'a': case 'e': case 'i': case 'o': case 'u': case 'A': case 'E': case 'I': case
    printf("%c is a vowel", ch);
    break;

default:
    printf("%c is a consonant", ch);
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE

```
cd "/Users/adarshaprasai/Documents/Herald/Final Year/Sem1/HPC/Bonusclass_cw/output"
./"largestofthree"
adarshaprasai@Adarshas-MacBook-Air Bonusclass_cw % cd "/Users/adarshaprasai/Documents/Herald/
adarshaprasai@Adarshas-MacBook-Air output % ./"largestofthree"
Enter number1:
10
Enter number2:
11
ENter number3:
3
The largest number is: 11%
adarshaprasai@Adarshas-MacBook-Air output % cd "/Users/adarshaprasai/Documents/Herald/Final Y
./"leapyear"
adarshaprasai@Adarshas-MacBook-Air output % ./"leapyear"
Enter a year:
2020
 Year 2020 is a leap year%
adarshaprasai@Adarshas-MacBook-Air output % cd "/Users/adarshaprasai/Documents/Herald/Final Y
./"vowelornot"
adarshaprasai@Adarshas-MacBook-Air output % ./"vowelornot"
Enter a character:
e
e is a vowel%
adarshaprasai@Adarshas-MacBook-Air output %
```

6. Star pattern

leapyear.c | largestofthree.c | vowelornot.c | sum_array.c

star.c > ...

```c
#include <stdio.h>
int main() {
    int rows, i, j, space;
    printf("Enter number of rows: ");
    scanf("%d", &rows);
    for (i = 1; i <= rows; i++) {
        for (space = 1; space <= rows - i; space++) printf(" ");
        for (j = 1; j <= (2*i - 1); j++) printf("*");
        printf("\n");
    }
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE

```
cd "/Users/adarshaprasai/Documents/Herald/Final Year/Sem1/HPC/Bonusclass_cw
./"star"
adarshaprasai@Adarshas-MacBook-Air Bonusclass_cw % cd "/Users/adarshaprasai
adarshaprasai@Adarshas-MacBook-Air output % ./"star"
Enter number of rows: 5
    *
   ***
  *****
 *******
*********
adarshaprasai@Adarshas-MacBook-Air output %
```

7. Calculate sum of array elements

leapyear.c | largestofthree.c | vowelornot.c | sum_array

sum_array.c > main()

```c
#include <stdio.h>
int main (){
    int arr[5], i, sum = 0, j;
    for(i = 0; i < 5; i++){
        printf("Enter number %d: \n", i+1);
        scanf("%d", &arr[i]);
    }
    for(j = 0; j < 5; j++){
        sum = sum + arr[j];
    }
    printf("The sum of the array is: %d", sum);
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE

```
cd "/Users/adarshaprasai/Documents/Herald/Final Year/Sem1/HPC/Bonusclas
./"sum_array"
adarshaprasai@Adarshas-MacBook-Air Bonusclass_cw % cd "/Users/adarshapr
adarshaprasai@Adarshas-MacBook-Air output % ./"sum_array"
Enter number 1:
1
Enter number 2:
1
Enter number 3:
1
Enter number 4:
1
Enter number 5:
1
The sum of the array is: 5%
adarshaprasai@Adarshas-MacBook-Air output %
```

8. Find length of a string (without using strlen)

leapyear.c | largestofthree.c | vowelornot.c | su

string_len.c > ...

```c
#include <stdio.h>
int main() {
    char str[100];
    int i = 0;
    printf("Enter a string: ");
    scanf("%s", str);
    while (str[i] != '\0') i++;
    printf("Length of the string: %d\n", i);
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE

```
cd "/Users/adarshaprasai/Documents/Herald/Final Year/Sem1/HPC/Bo
./"string_len"
adarshaprasai@Adarshas-MacBook-Air Bonusclass_cw % cd "/Users/ad
adarshaprasai@Adarshas-MacBook-Air output % ./"string_len"
Enter a string: adarsha
Length of the string: 7
adarshaprasai@Adarshas-MacBook-Air output %
```

9. Reverse a string

leapyear.c | largestofthree.c | vowelornot.c

string_reverse.c > ...

```c
int main() {
    int i = 0, j, temp;
    printf("Enter a string: ");
    scanf("%s", str);
    while (str[i] != '\0') i++;
    j = i - 1;
    for (i = 0; i < j; i++, j--) {
        temp = str[i];
        str[i] = str[j];
        str[j] = temp;
    }
    printf("Reversed string: %s\n", str);
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS A

```
cd "/Users/adarshaprasai/Documents/Herald/Final Year/Sem1/HP
./"string_reverse"
adarshaprasai@Adarshas-MacBook-Air Bonusclass_cw % cd "/Users
adarshaprasai@Adarshas-MacBook-Air output % ./"string_reverse
Enter a string: adarsha
Reversed string: ahsrada
adarshaprasai@Adarshas-MacBook-Air output %
```

10. Prime number check using function

leapyear.c | largestofthree.c | vowelornot.c | sum_array.c | string_len.c

prime.c > ...

```c
#include <stdio.h>
int isPrime(int num) {
    if (num <= 1) return 0;
    for (int i = 2; i <= num/2; i++) {
        if (num % i == 0) return 0;
    }
    return 1;
}
int main() {
    int n;
    printf("Enter a number: ");
    scanf("%d", &n);
    if (isPrime(n)) printf("%d is a prime number.\n", n);
    else printf("%d is not a prime number.\n", n);
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE

```
cd "/Users/adarshaprasai/Documents/Herald/Final Year/Sem1/HPC/Bonusclass_cw/output"
./"prime"
adarshaprasai@Adarshas-MacBook-Air Bonusclass_cw % cd "/Users/adarshaprasai/Documents/Herald/Final
adarshaprasai@Adarshas-MacBook-Air output % ./"prime"
Enter a number: 51
51 is not a prime number.
adarshaprasai@Adarshas-MacBook-Air output %
```